

Bullet Farm: Architecture and Pre-Release Inventory of a Software-Change Transaction Processor

Bullet Farm Maintainers

Abstract—Coding agents are proliferating as chat loops, plugin kernels, and meta-wrappers around existing CLIs. The scarce resource is not another persona or sandbox flag. It is a correct concurrent *software-change transaction*: who may mutate an exact repository subject, which incarnation owns that write, what evidence names that subject, and whether an ambiguous remote effect may be retried. Bullet Farm is a local-first transaction processor for software changes and model cognition. Models may propose typed patches. They cannot grant write, verification, or merge authority. This preprint inventories the architecture and the exact component receipts that exist on 2026-08-25, states a threat model and an explicit non-claim list, and compares that contract with Gas Town / Gas City, DeepSeek Harness, Databricks Omnigent, and the broader coding-agent landscape. Many Kernel, BulletGit, and Portal machines now have commit-bound component proofs, including signed launch grants and live-tested Linux egress isolation. The connected five-plane product and every family release gate remain blocked. The paper claims a stronger systems contract, not measured superiority.

Index Terms—Multi-agent software engineering, transaction processing, Git isolation, evidence binding, authority fencing, coding agents, verification.

I. INTRODUCTION

The last two years produced a crowded market of coding agents: protocol-grade CLIs, IDE wrappers, hosted pull-request bots, research SWE loops, multi-agent orchestrators, plugin kernels, and meta-harnesses that wrap the previous categories [1]–[7]. Those systems differ in user experience and in how much of the model loop they own. They largely agree on an implicit completion rule: if a session looks done, a terminal is idle, a process exited zero, or a pull request opened, the work is complete.

That rule fails under concurrency. Public Gas Town and Gas City issues show the same family of faults repeating under load: two agents share one working tree and clobber uncommitted edits [8]; a timeout after a server-side commit mints duplicate live workflows and three writers enter one checkout [9]; a background `git checkout` against the town-root `.git` deletes operational files [10]; a dead `tmux` session is treated as live ownership and drives hundreds of respawns [11]. Those reports are not a verdict that Gas Town is a weak product. They are evidence that a role/session orchestrator is the wrong abstraction for adversarial multi-writer Git [12].

DeepSeek Harness and Omnigent improve different layers. DeepSeek treats every capability—model, tools, sandbox, session, and UI—as a Cordis plugin and makes the append-only

session log the source of model-visible context [13], [14]. Omnigent sits *above* Claude Code, Codex, Cursor, and other CLIs, adding OS sandboxes, an L7 credential proxy, contextual policies, and live collaboration [7], [15], [16]. Both are real advances. Neither binds write authority to an incarnation-fenced Attempt, evidence to an exact Candidate, or remote integration to a reconciled effect receipt. A plugin swap or a wrapped harness can still let the model run `git`.

Bullet Farm inverts the authority model [17], [18]. The designed law is compact:

Many agents may reason. Multiple isolated Variants may compete. Exactly one incarnation-fenced Attempt may write within each Variant. Every consequential external mutation is brokered. Every proof names its exact subject. The repository remains sovereign.

The contribution of this preprint is fivefold. First, it states that law as a five-plane transaction, not as a prompt convention (Section IV). Second, it shows why Git itself is the deepest missing substrate and how BulletGit answers a closed list of Git failure classes (Sections III and VII). Third, it reports *exactly* what is implemented as of the later 2026-08-25 closure reconciliation—component receipts versus blocked product gates—so a reader cannot confuse a green crate test with a release (Section VIII). Fourth, it states a threat model, the Centerrail C1–C12 disposition, a non-claim list, and a red-team self-review so a hostile reader has the attacks we already accept (Sections V, VI, and XIII, Appendix B). Fifth, it enumerates every remaining V1 product gap so an operator cannot treat missing prose as missing definition (Appendix C). Tables I and IV compare authority and Git bindings. Table V maps each failure class to a rule and to an honest implementation status.

This paper does not claim that Bullet Farm is faster, cheaper, or more autonomous than any named system. No public comparison is valid until the same receipt-bearing corpus runs on pinned upstream commits [19]. The previous separately produced IEEE draft recorded that Bullet Farm had not been implemented [20]. That sentence is now false for many components and still true for the connected product.

II. A TAXONOMY OF CODING AGENTS

Treating “coding agents” as one category hides the authority question. We distinguish six families by *who writes the repository* and *what signal is allowed to mean done*.

A. Protocol-Grade Single Agents

Claude Code (bidirectional stream JSON), Codex (App Server JSON-RPC over stdio JSONL), Cursor (Agent Client

Placeholder authorship for this preprint. Named authors will be supplied before an arXiv upload. Affiliation: the Bullet Farm split family (public hub, kernel, BulletGit, and operations portal). This document is not a release receipt, installer, or benchmark result.

Protocol), and Google Antigravity (agy structured headless mode) are the four V1 adapter targets [1]–[4], [18], [21]. On 2026-08-25, sanitized local probes observed Claude Code 2.1.243, Codex CLI 0.149.1, Cursor Agent 2026.08.11-e8db854, and agy 1.1.19. Those observations are inventory, not certification. Cursor’s required ACP entry point remained UNKNOWN on that host; print/stream-JSON modes must not be substituted [18]. Each of these products is a capable single-agent harness. The model, or a tool the model invoked, writes the workspace. A “plan” flag or a vendor sandbox is defense in depth, not a foreign write gateway.

B. IDE and CLI Wrappers

Aider, Cline, Continue, Goose, and Roo Code wrap an editor or terminal around one or more models [22]–[26]. They differ in UX, MCP adoption [27], and how much Git they automate. They share the user’s checkout as the writable world. Isolation, if any, is a worktree, a subdirectory, or an honor-system ignore list.

C. Research SWE Loops

OpenHands, SWE-agent, Agentless, and AutoCodeRover optimize for SWE-bench-style issue resolution [28]–[32]. Their contribution is the agent–computer interface and the evaluation harness. Completion is a resolved issue or a passing hidden test, not a protected integration onto a sovereign default branch under an independent attestor.

D. Hosted Change Agents

GitHub Copilot’s coding agent, Jules, Devin, and Kiro accept a task and return a pull request or a hosted workspace [33]–[36]. The PR is the product. That is a legitimate remote-collaboration contract. It is not an incarnation-fenced local transaction: retries, reviewer independence, and evidence staleness after rebase are delegated to the host’s merge queue and to human review.

E. Orchestrators

Gas Town is a Go workspace manager for many Claude, Codex, Copilot, Gemini, or Cursor agents [5], [37]. It persists work in git-backed *hooks* (worktrees), coordinates via Beads/Dolt, requires tmux 3.0+ for *gt up*, and lands work through a Bors-style Refinery [38]. Roles—Mayor, Witness, Deacon, polecat, Refinery—are operationally legible. Additional product machinery is easy to underestimate: convoys bundle beads; molecules instantiate TOML formulas; a three-tier watchdog (Witness, Deacon, Dogs) patrols health; a scheduler caps polecat dispatch to avoid provider 429s; Seance recovers predecessor session logs; Wasteland federates towns over DoltHub [5]. Gas City generalizes the same factory into Agent, Bead, Formula, Rig, Pack, and Event, with typed run/session APIs, warm projections, formula-v2 retry/fan-out/drain controls, degraded storage outcomes, and release artifacts with checksums, SBOMs, and attestations in v1.4.1 [19], [39], [40]. The product leverage is real. The authority model is not: liveness is a terminal multiplexer,

isolation is a worktree or an optional pack script, and the fleet can be the merge authority. Bullet Farm should be measured against those product properties—durable work, role flexibility, fan-out receipts, provider plurality, health reconciliation, and release evidence—not against a cartoon of “tmux and vibes” [19].

F. Plugin Kernels

DeepSeek Harness (dsh) is a developer-preview agent runtime whose doctrine is “everything is a plugin” [6], [14]. Cordis mounts model adapters, tools, sandboxes, storage, loops, scheduling, and the UI into one context [13], [41]. A running instance is a plugin tree composed from bundles and ordered patches. The session log is the source of model-visible context: anything the model sees must be reconstructable from that log. A turn is zero or more steps; each step is one model request plus the tools it calls; agent/pre-step may rewrite or reject claimed messages; tool execution is a pre-execute/execute/post-execute waterfall [13]. Standard mode gives the model a file editor, a shell, search, skills, and subagents; Code mode lets the model orchestrate tool rounds in TypeScript; Minimal mode keeps bash and *str_replace_editor*; Creator mode inspects the runtime to author presets [14]. Experimental Agent Teams add a roster, task board, and mailbox over continuable subagents. This is an excellent composition kernel for a *single* agent product. It is not a transaction kernel. Filesystem and subprocess providers share one execution world, so pointing them at a remote sandbox moves Bash and PTY together—and the model still writes files. A durable session event is not a Candidate.

G. Meta-Harnesses

Omnigent is an open-source meta-harness, with a Databricks-operated deployment, that wraps Claude Code, Codex, Cursor, OpenCode, Hermes, Pi, Antigravity, and custom YAML agents [7], [15], [42]. A runner exposes a uniform session API. A server stores policies and sharing. Native Linux wrappers use bubblewrap; macOS uses seatbelt; an L7 egress proxy can attach credentials so the sandbox never holds the real token [16]. That credential-proxy design is one of the strongest OS-isolation stories in this survey. Native terminal wrappers still require tmux. The wrapped harness still performs Git. A policy can deny, transform, or pause a tool call; it cannot mint a Candidate or invalidate evidence when the branch moves.

III. RECURRING FAILURE CLASSES

The issue corpus and the Git documentation converge on a small set of missing *bindings* [12], [43], [44]. A binding is a rule that makes a class of silent success impossible. Role names, dashboards, and sandboxes do not substitute. Table I summarizes how the surveyed systems bind—or fail to bind—write, completion, evidence, integration, and liveness.

TABLE I
AUTHORITY BINDINGS. “TYPICAL HARNESS” IS THE CLAUDE CODE / CODEX / CURSOR / ANTIGRAVITY SINGLE-AGENT PATTERN [18].

System	Who writes the repo	Isolation	Completion signal	Evidence subject	Integration authority	Liveness oracle
Bullet Farm (designed)	bullet-gitd only	Private clone, no worktree	Candidate + independent Evidence + forge receipt + observation	{base, head, tree, patch, env}	Jeryu/GitHub required checks; attestor $\neq$ broker	Lease/fence; ALIVE/DEAD/UNKNOWN/CC
Bullet (implemented)	Farm Production refused write	Dissociate clone code exists; not reachable online	PENDING→UNKNOWN/FAILURE only	One fixture E2 gate; no live subject	Local bare-forge compo- nent; Jeryu quarantined	DB-clock leases; no signed BulletGit online final-authority mint tmux session
Gas Town v1.2.1	Agent in hook worktree	Git worktree sharing .git	gt done, convoy, Re- finery merge	Gates exist; not exact- OID bound	Refinery / local merge queue	
Gas City v1.4.1	Agent; implicit pool uses rig root [8]	Pack-optional work- tree	Formula/run/session closure	Session and bead state	Configurable; still orchestrator-owned	Session + controller
DeepSeek Harness rc.2	Model via fs/shell plu- gins	Sandbox plugin back- end	Turn/session log	Session events (model- visible)	n/a (not a forge OS)	Agent loop
Omnigent v0.10.0	Wrapped harness	OS sandbox + L7 proxy	Session / policy deci- sion	Policy and session events	n/a (meta-harness)	tmux on native wrap- pers
Typical harness	single-agent The agent in the user checkout	Vendor sandbox flags (optional)	Model “done”, PR, or exit 0	Transcript or CI on a branch	Hosted PR or human merge	Process / IDE session

A. Shared Writable Identity

Git worktrees share one object store, one set of refs, and—unless extra machinery exists—one index and one hook namespace [43]. Gas City issue 1181 records implicit pool agents resolving every session to the rig root: two parallel slings clobber uncommitted edits [8]. Gas Town issue 4094 records background polecat operations running `git checkout` against the town-root `.git`, deleting untracked operational files such as `mayor/rigs.json` and Dolt manifests [10]. A worktree is a checkout convenience. It is not a private repository.

`git clone -reference without -dissociate` writes `objects/info/alternates` into a borrowed store. A later `git gc -prune=now` on the mirror can unlink objects a live workspace still names [20], [43]. CoW reflinks without a documented fallback have the same ownership question.

B. Hostile Configuration and Inherited Credentials

A repository-local or inherited Git config can install filters, aliases, include files, URL rewrites, credential helpers, hooks, and `protocol.file` transports that execute host code or exfiltrate bytes during an ordinary `checkout` or `diff` [43], [45]. Case-folded path collisions on a case-insensitive volume, alternate data streams, and symlink traversal are the same class: Git will write something the operator did not grant. Model-authored `git` in a process that inherited `GIT_*`, `SSH_AUTH_SOCK`, or a forge token is a credential and integrity failure, not a convenience. Compile-time code—Rust `build.rs`, `proc-macros`, TypeScript transformers—executes before any test “runs” [20]. Sandboxing only the test process is insufficient. Detached `HEAD`, an active sequencer (`MERGE_HEAD`, `REBASE_HEAD`), and a dirty index shared across worktrees convert ordinary “just commit” steps into lost or mis-attributed work [43].

C. Control Plane in Git

Git is a poor coordination database for renewable leases, heartbeats, quota reservations, and unsettled remote effects [44], [46]. Beads-on-Dolt-in-Git is an inventive store and

an operationally heavy one: install depends on Dolt, ICU, and `bd`, and writer-deadlock classes appear under load [5], [9]. Putting the live ledger in the same bytes the agents edit couples control-plane availability to repository mutation.

D. Inferred Completion and Unsafe Liveness

Gas Town’s problems-view calls a dead agent a “Zombie: Dead tmux session” [5]. Issue 742 shows the failure: a dead session left work claimed and drove 687 respawns in 24 h [11]. A timeout after Dolt committed server-side is classified transient and retried, yielding three complete workflows from one `gc sling` [9]. Hosted agents treat PR-opened as done. Plugin kernels treat turn-end as done. In every case, transport success is painted as domain success.

E. Evidence Not Bound to an Exact Subject

A green CI result that names a branch, a worktree path, or a session ID is not evidence about a rebase, a repair commit, or a merge-group composition. Ordinary Git commits have no stable Change identity across rewrite [43], [47]. After the branch moves, yesterday’s green is a different subject. Merge queues that do not invalidate overlapping evidence create silent stale-green admission; queues that invalidate every queued candidate recreate a livelock when arrival rate exceeds verification time [20].

F. Destruction Without Preservation

Cleanup, GC, polecat reuse, and “nuke” operations that target a name, path, or inferred age destroy the only copy of uncommitted work [10], [12]. Preservation must bind an exact incarnation and a sealed artifact *before* deletion eligibility.

IV. BULLET FARM ARCHITECTURE

Bullet Farm is a split family: a public hub (`bullet-farm`), a control-plane kernel, a Git capability daemon, and a projection-only portal [17]. Jeryu is the source forge and the mandatory local effect adapter. GitHub is a separately configured effect adapter, never source authority.

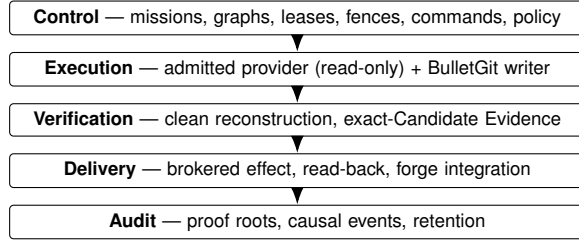


Fig. 1. Five trust planes. A plane may consume only grants addressed to its principal. Portal and any evolutionary optimizer are projections or requestors, never mutation credentials [49].

A. The Software-Change Transaction

Mission → GraphRevision → Variant → Attempt
 → Proposal → Candidate → Evidence (1)
 → Effect → Integration → Observation.

A Mission is a durable objective plus constraints. A Graph Revision is an immutable package decomposition. A Variant is one hypothesis with an immutable role, routing, context, and configuration snapshot. An Attempt is one execution incarnation under a unique lease and a never-reused fence. A Proposal is typed patch data. A Candidate is a provenance-bound immutable phenotype created only by BulletGit. Evidence is an independently reproduced result over that exact Candidate and admitted gate definitions. An Effect is a brokered remote mutation with read-back. Integration is protected-ref admission by the forge. Observation is post-merge survival. None of those words is a prompt label. Each is a typed subject with an exact identifier [17], [48].

B. Five Trust Planes

ADR 0003 separates control, execution/repository write, independent verification, effect/attestation/integration, and observation/audit [49]. Compromise of the portal, a provider, or a broker must not manufacture independent evidence once those principals exist as distinct identities. ADR 0003 *requires* distinct OS identities on one host and SPIFFE-compatible identities later; that separation is not a current receipt. Production processes still share a UID, and same-UID path-based Git races remain an open setup boundary [50]. The evolutionary control document defines roles as capability envelopes—Planner, Implementer, Critic, Verifier, Effect broker—not as personas that carry credentials [48]. Online learning, larger councils, and distributed execution are Phase 9–10 and are out of V1 [48].

C. Three Sovereign Truths

A common instinct is to put missions, leases, and evidence into Git because the product is about Git. That collapses two badly matched stores [44]. Bullet Farm splits canonical truth:

- **Kernel ledger** (SQLite WAL today; PostgreSQL is a refused scaffold [51]) owns mutable authority: leases, fences, commands, policy snapshots, outbox, event log.

- **BulletGit** owns immutable engineering subjects: Changes, Candidates, checkpoints, journals, proof roots.
- **Forge** (Jeryu, optionally GitHub) owns remote refs, required checks, and the merged target SHA.

Cross-boundary values come from generated bullet-wire contracts: RFC 8785 canonical JSON, domain-separated BLAKE3, full 256-bit IDs, algorithm-tagged Git OIDs, and PASETO v4.public envelopes [52]–[54]. Prose and TypeScript types do not define the wire.

D. Providers Propose

ADR 0001 is the execution-mode decision [18]. Every provider turn is read-only inside a private workspace generation. The only write input is a validated `PatchProposal`: schema identity, producing `Attempt`, base checkpoint, per-path preimages, and policy-admitted `gate_ids`. Provider text such as `tests_to_run` or a shell command is never argv. Runner resolves gate IDs through a trusted catalog. Zero tests, timeout, skipped, flaky, unsupported, and UNKNOWN never equal PASS. At most two bounded repair turns may see typed gate results.

Public mutations use authenticated
 POST /v1/commands, initially return
 202 PENDING, and reconcile through
 PENDING|APPLIED|VERIFIED|FAILED|UNKNOWN [54].
 Transport success never implies verification. The Portal renders only those phases and treats watermark mismatch, sequence gaps, and timeouts as UNKNOWN or STALE, never as an empty success list [55].

E. Why the Bindings Are the Product

Other systems can copy a role name in a week. They cannot copy a binding without changing what an agent is allowed to *be*. A Mayor that still infers liveness from tmux is a persona. A fence that is checked inside the ledger transaction at the effects gateway is a binding. A plugin that swaps the filesystem provider is composition. A daemon that is the only process allowed to apply a scoped patch is a binding. A contextual policy that pauses a tool call is governance. An Evidence record that becomes invalid when `head` moves is a binding.

That is the source of leverage. As models improve, a harness that lets the model write Git gets more destructive more quickly. A transaction processor that lets the model propose gets more useful more quickly, because the same fence, Candidate, and verifier contracts apply to a weaker economy model and to a frontier council. The evolutionary document states the same rule for search: hard constraints are filters, never terms a high score may offset, and the optimizer cannot see or learn away the safety envelope [48]. V1 records routing provenance and does not learn online.

F. A Worked #5511-Class Fault

Issue 5511 is the canonical timeout-after-commit race [9], [20]. Under load, `bd create -graph` is killed at the runner deadline after Dolt has already committed. The client

sees a timeout, classifies it transient, and re-applies. The result is three live workflows and three writers in one checkout.

Bullet Farm’s designed handling is mechanical:

- 1) Graph materialization is one serializable insert keyed by content hash. A replay returns the existing graph; it does not mint a sibling.
- 2) The command that requested materialization is recorded with an idempotency key. An ambiguous timeout becomes UNKNOWN, not a retry.
- 3) A resolver may adopt a remote or ledger subject only when its identity equals this Attempt’s desired state. Otherwise the subject is ORPHANED and quarantined.
- 4) Even if two Attempts exist, each requires its own never-reused fence and its own private clone. They cannot share a cwd or a .git.
- 5) A writer that outlives its lease cannot push, comment, or settle: the fence is presented at the effects gateway and rejected inside the ledger transaction.

Items 2 and the lease/fence machine have component proofs. Item 1 is specified and is not a live path at HEAD. The point of the walkthrough is not to claim the incident is closed in production. It is to show that the missing primitive is identity-exact adoption under ambiguity, not a smarter Mayor prompt.

G. Evidence Tiers

Writer output cannot satisfy an independent tier [18], [48]. The designed ladder is:

- E0 — untrusted provider or author claim;
- E1 — writer-visible gates on the author’s workspace;
- E2 — clean independent reconstruction of the exact Candidate;
- E3 — hidden or historical holdout, plus hostile suites the author cannot weaken;
- E4 — human review.

The implemented verifier produces typed E2-shaped results for one fixture gate over exact base/head/tree subjects [54]. Multi-gate aggregation, executable-digest admission, and E3 holdout custody remain open. That is the correct partial result: the identity rule exists before the gauntlet is populated.

V. THREAT MODEL

We assume a single-user, loopback V1 host [54]. The adversary is not a nation-state on the wire. The adversary is (i) a frontier model that will try to run shell, Git, and forge tools; (ii) a second concurrent Attempt that shares a name, path, or session identity; (iii) a crashed or partitioned runner that retries after an ambiguous timeout; (iv) hostile repository content (hooks, filters, build.rs, proc-macros, .gitattributes); (v) a compromised or confused portal that wants to paint UNKNOWN green; and (vi) an operator who would rather flip an environment variable than mint a lease-bound grant.

Out of scope for V1, and therefore not silently promised: multi-tenant SaaS, remote untrusted runners, cross-repository sagas, online routing learners, and a formal proof that physical side effects are exactly-once. Those absences are named so a red team cannot treat them as implied.

Trusted computing base for an admitted write is the Kernel ledger transaction, the signed launch-grant verifier, the egress-isolation receipt, bullet-gitd’s fail-closed checker, and the forge’s required checks. The model, the Portal, MCP servers, vendor sandboxes, and tmux are untrusted. A valid launch grant binds the durable active lease, the evaluated provider admission (absolute path and digest), the policy snapshot, and a ≤ 15 s single-use nonce; it never authorizes lease acquisition, Attempt advance, repository mutation, or effects [56]. Today’s committed policy still refuses live admission, so a fully valid grant ends in POLICY_LIVE_ADMISSION_DISABLED without spending the nonce [57]. A same-UID pathname swap between lock verification and Cargo/Node/npm exec is an accepted residual of source-tree setup until a signed prebuilt installer exists (G1). It is not a silent implication that hub-only install is safe.

We also treat as in-scope, and currently unmitigated at HEAD: same-UID swap of a path-based Git or installer subject [50]; prompt-shaped instruction files the provider will read before any PatchProposal exists [58]; a compromised Kernel ledger or bullet-gitd process (either is total loss on a single-user host); and a quarantined backup restore replayed as production authority [50]. Distinct OS identities, signed internal lease transport, and a Kernel loader that understands v1alpha2 are designed counters, not receipts. The two TLA+ models are locked state-enumeration artifacts (TypeOK plus local invariants; deadlock checking is off). They are not a third product gate and they are not a deadlock-freedom proof [20], [59].

VI. RED-TEAM DISPOSITION (C1–C12)

The family TEAM.md document is a hostile review of the Centerrail specification [60]. It is historical provenance, not runtime authority. Living control is the evolutionary document, the release contract, and the product-gap register [48], [50], [61]. We adopt the twelve critiques that simplify or harden with no regression, and we keep the four rejected cuts rejected: the verifier plane stays; SQLite remains the V1 ledger; the five planes stay; the forge remains merge-sovereign.

Table II is the product status of those twelve hardenings. “Adopted” means the rule is the current design. “Component” means a named commit and mapped tests exist. “Designed” means the rule is specified and not a live path. A designed row is not a silent implication that the tree enforces it.

C7 is the one place the original build plan over-reached and the tree already followed the fix: exactly two TLA+ models, not seven, before Phase A code [59]. C8 is why four adapters are a goal and not a GA hostage. C2 and C9 remain the load-bearing unshipped hardenings: without an oracle the author cannot weaken, and without identity-exact adoption, a timeout-after-commit still launders a stale write. Those two are why a connected TRANSACTION_PROOF is the next product, not a larger council [61].

VII. BULLETGIT: THE MISSING SUBSTRATE

Models and harnesses commoditize. An agent-first source-control substrate that understands authority, change identity,

TABLE II

CENTERRAIL C1–C12 DISPOSITION AGAINST HEAD. ADOPTED IS DESIGN LAW. COMPONENT IS A RECEIPT. DESIGNED IS SPECIFIED AND NOT SHIPPED.

ID	Adopted meaning	Product status
C1	Every invariant is T1 schema / T2 gateway / T3 test	Registry exists; most later-wave rows remain planned
C2	Oracle-modifying diffs; holdouts required for R2+	Designed; implemented verifier is one fixture E2
C3	Two-track scope expansion [62]	Accepted ADR; not a live Attempt path
C4	Attestor $\neq$ broker; reconstructible check	Designed; live forge blocked
C5	CONTRADICTION has fence-mediated exits	Designed
C6	Bounded probe reservation; unknown $\neq$ headroom	Designed
C6b	One seat-equivalent per named human	Designed; not enforced in the kernel
C7	Formal-model exactly two Phase-0 protocols	Adopted and component-complete
C8	GA = kernel + any two certified providers	Adopted as schedule; live still operator-gated
C9	Identity-exact adoption (fence + desired OID)	Command idempotency component; graph mint not live
C10	Verifier dwell back-pressures writers	Designed
C11	Freeze chip: recorded vs enforced-on- N/M	Portal honesty component; countdown designed
C12	Saga quarantines blast radius, not the fleet	Out of V1

proof, lineage, and protected integration does not [44]. Bullet-Git is that substrate. `bullet-gitd` is the sole writer of the private workspace [45]. Pack protocol, refs, and protected updates remain ordinary Git at the forge.

A. Private Clone, Not a Worktree

Workspaces never clone the operator’s source tree directly. A bare mirror is created or fetched under an exclusive lock. The private clone runs

```
git clone -reference-if-able <mirror>
      -dissociate
```

while the lock is held [45]. Objects may be borrowed during the copy; no alternates file survives. Mirror GC cannot corrupt a live workspace. Immediately after clone, `origin` is removed, `credential.helper` is emptied, `GIT_ASKPASS` points at a deny script, and `GIT_SSH_COMMAND=false`. A model-issued push has no destination and no authenticator.

A `.git file`—the on-disk shape of a worktree—is `WORKTREE_FORBIDDEN`. Sequencer files at checkpoint time are `SEQUENCER_ACTIVE`. Checkpoints run `git write-tree` against a temporary `GIT_INDEX_FILE`, never the live index.

B. Hostile-Git Refusal

The child environment is cleared of inherited `GIT_*` variables and rebuilt with per-workspace `HOME/XDG`, `GIT_CONFIG_NOSYSTEM=1`, and `GIT_CONFIG_GLOBAL=/dev/null` [45]. Every invocation disables paging, hooks, credentials, filesystem monitors, and external attribute files, and passes `protocol.file.allow=never` except for the one local clone that needs file transport. Before a repository-scoped command, `BulletGit` parses local config with includes disabled. Command-bearing or truth-redirecting filter, alias, include, pager, URL rewrite, credential, hook, or worktree settings fail closed as `HOSTILE_GIT_CONFIG`. A planted clean filter behind `.gitattributes` is a required refusal test.

C. Scope, Identity, Preservation

`apply_change` is specified to validate every path against a `ScopeGrant` before writing: Unicode NFC, no `...`, no `.git` component, no alternate data streams, no symlink traversal. Duplicate and case-folded collisions must refuse the whole batch. Wire-shaped `CandidateId` bytes exist; a provenance-complete `Candidate` identity (distinct `ContentId`, complete proof manifests) is still open at the inventory HEADs [45], [54]. `patch_hash` is specified as BLAKE3 of `git diff base...head`. Local cleanup accepts only an opaque preservation receipt that re-verifies the sealed generation immediately before deleting exactly those bytes; online settlement is not a live path [45].

D. Honest Production Boundary

The public daemon constructor installs an unavailable final-authority checker. In a fresh daemon, `clone` returns `AUTHORITY_CONTRACT_UNAVAILABLE` before repository I/O [45]. Later verbs fail `NOT_CLONED`. The unsigned legacy token grants nothing. Production enablement still requires a published immutable `bullet-wire` tag, local PASETO verification, Kernel online reservation, and one-second operation permits. That fail-closed posture is a feature of the current tree, not a footnote.

VIII. EXACT INVENTORY: WHAT WORKS TODAY

Table III is a receipt table, not a feature list. Subjects are those recorded by the later 2026-08-25 closure reconciliation [50], [54], [61]. Family HEADs at that reconciliation were `hub bf5c64245d218050eb8f4c3228ea15654540a94e`, `kernel 0346fd531e231a88e5b1eb3a8257be04f235068b`, `BulletGit 236f4efe2b8243b3c4dd5da13269e45119b061df`, and `portal 8272844e44d89ea3b26aa5b3c4f25909a069e101`. Those commits include the operator-ratifiable `v1alpha2` validator and the synchronized schema-bundle consumers. The portal bundle root at that subject is `blake3:556d91f8504299230bc75e027108d1d943fd9ff9930`. A later commit invalidates a row until the mapped gate is replayed. This paper does not silently track moving tips. Earlier preprint drafts named `hub 74507b30dd40` / `kernel ba70d8ba7bef` / `BulletGit 5dac98e12c1a` / `portal d607068901a7`; those are superseded as inventory subjects.

TABLE III
SELECTED COMPONENT RECEIPTS VERSUS PRODUCT GATES
(2026-08-25). COMPLETE MEANS A NAMED COMMIT AND MAPPED TESTS
EXIST. IT DOES NOT MEAN A RELEASE.

Subject	Class	Boundary
Signed wire DTOs, JCS, PASETO shapes	COMPLETE	No running issuer; no immutable published tag
DB-clock leases, TTL 1..15s, never-reused fence	COMPLETE	No signed final-authority mint
Atomic lease/command/event/outbox	COMPLETE	Public dispatch of providers/effects open
Four offline provider parsers (Claude, Codex, Cursor, AntigraVity)	COMPLETE	Public spawn refused; no live conformance
Admitted fixture gate + bounded verifier transport	COMPLETE	One fixture gate; no JSON-RPC production path
Fail-closed bullet-gitd + dissociate clone + hostile-git + preservation	COMPLETE	Production clone refused
Portal CSRF/202 + PENDING→UNKNOWN + SSE STALE	COMPLETE	Not packaged/embedded; 11 of 15 catalog surfaces stay explicit UNKNOWN
Setup refuses schema-2 hub-only install	COMPLETE	No signed schema-3 lock or prebuilt installer
Two TLA+ models (lease/fence; effect ambiguity)	model lock	TypeOK plus local invariants; deadlock checking is off [59]
Signed launch grant + Linux egress isolation	COMPLETE	Live dispatch policy-disabled; no HTTP mint [56]
Policy v1alpha2 validator (hub)	COMPLETE	ADR 0012 Proposed; Kernel loader still v1alpha1-only; no committed policy enables live admission [57]
Five-plane TRANSACTION_PROOF	BLOCKED	No signed cross-plane crash receipt at HEAD
Live provider / Jeryu / GitHub effects	BLOCKED	Operator ratification and forge credentials
Hub Jankurai ≥ 90 , zero hard findings	BLOCKED	Score 58; 26 hard findings
Family check release	BLOCKED	26/26 gates; static negative inventory
Evolutionary recipe archive	designed	Gated on a working transaction; not started

Kernel architecture is equally explicit about what the bins are not [51]. `bullet-farmd` records PENDING and an internal reconciler can settle known demo work only to UNKNOWN or unsupported kinds to FAILED. It cannot emit APPLIED or VERIFIED. Provider admission compares an absolute executable to a BLAKE3 digest, builds a unique mode-0700 HOME, copies only digest-checked OAuth files as mode 0400, and strips host PATH, SCM, cloud, SSH, and API secrets. Signed launch grants and live-tested Linux user+net namespace egress now exist as components [56]. Dispatch remains policy-disabled. A fully valid launch grant under the committed generation-1 `v1alpha1` snapshot ends in `POLICY_LIVE_ADMISSION_DISABLED` without spending the nonce [56], [57]. Hub commit `bf5c642` accepts a `v1alpha2` snapshot on paper only when `policy_generation` ≥ 2 and an unrevoked provider-runner PASETO key is registered; the Kernel policy loader still accepts `v1alpha1` only, so a `v1alpha2` file is not an admission path [50], [57]. The four provider crates remain transcript machines until that operator ratification and a loader mirror exist. Feature-gated tests that pass by refusing spawn are quarantine evidence, not conformance.

Unauthenticated farmd lease HTTP is not an admission path and must not be remounted to mint a grant [56].

A launch grant is a one-second-to-fifteen-second capability, not a session cookie. In the designed and now component-implemented shape it is a PASETO v4.public token whose claims bind the durable active lease (Mission, repository, graph revision, work package, Variant, Attempt, fence, runner epoch, workspace nonce), the evaluated admission (absolute executable path and BLAKE3 digest, profile, protocol), the policy snapshot digest, and a single-use nonce [56]. Audience is `provider-runner`; operation is `launch-provider`. Minting is in-process from an operator-held mode-0600 key. There is no HTTP mint route. Clearing `SIGNED_ADMISSION_UNAVAILABLE` does not clear `EGRESS_ISOLATION_UNAVAILABLE`; those are independent blockers. Clearing both still does not dispatch under the committed generation-1 policy [57]. That stack is the opposite of `BULLET_LIVE_PROVIDERS=1`: an environment variable is not admission and was rejected [56].

An archived live demo receipt from 2026-08-24 14:04Z exists in the family evidence bundle. HEAD policy keeps `live_admission_enabled=false`. That archive does not certify the current tree and does not clear a release gate [50].

Exactly two protocols are model-checked in TLA+ and pinned by a model lock: lease/fence/reclaim, and command/effect ambiguity under timeout [54], [59]. A third formal model is not a V1 gate. The implemented models must continue to encode fence monotonicity; a skeleton that omits the conflict path does not earn a deadlock-freedom claim [20].

The citable facts are the receipt rows and the blocked gates, not an internal weighted score.

IX. COMPARISON

Tables I and IV use pinned subjects: Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness `dsh-v0.1.1-rc.2` (b150a55), Omnigent v0.10.0 (40755dd) [6], [19], [37], [39], [42]. Empty knowledge is written unknown, never guessed. Bullet Farm is split into the designed contract and the implemented tree so a reader cannot promote a document into a daemon.

The comparison is deliberately unfair to Bullet Farm on maturity and fair to everyone on bindings. Gas Town is a shipped orchestrator with substantial operational texture. DeepSeek is a composable agent product. Omnigent is a collaboration and governance layer that can wrap the others tomorrow. Bullet Farm is pre-release. What the others do not have—and what makes the architecture categorically different—is the set of bindings in the first row of each table. A system can add a Mayor, a plugin, or a policy function and still retry an UNKNOWN push into a shared worktree.

A. What We Must Not Lose From Each System

Gas Town’s factory metaphor is the right product skin for a fleet: named work, visible roles, escalation, and a merge rail an operator can watch [5], [12]. Bullet Farm should keep those surfaces as projections over the event log, not as

TABLE IV
GIT AND WORKSPACE SUBSTRATE. “IMPL.” IS THE BULLET FARM TREE AT THE SECTION VIII SUBJECTS.

System	Private clone	Dissociate / CoW	Hostile-Git refuse	Path scope grant	Stable Change/Candidate	Preserve-before-destroy
Bullet Farm designed	required	required	required	required	required	required
Bullet Farm impl.	code present; production clone refused	-reference-if-ab -dissociate implemented	implemented in SafeGit	implemented	wire-shaped IDs; provenance-complete Candidate still open	implemented locally; online settlement open
Gas Town / City	no (workspace / shared cwd)	not the isolation story	unknown as a kernel	optional pack scripts	bead IDs, not Candidate OIDs	reuse/reset; issue 4094 class
DeepSeek Harness	unknown (workspace = fs provider)	n/a	plugin policy, not Git parser	tool/sandbox policy	session IDs	session persist, not sealed generation
Omnigent	sandbox FS, not a private .git	n/a	OS + credential proxy	policy handler	session IDs	sandbox lifetime
Typical harness	user's checkout	rare / foot-gun if used	inherited Git config	prompt or ignore file	Git commit SHA only	leftover branches

authority-bearing model personas. Gas City’s primitive test—keep judgment out of transport, prefer primitives that remain useful as models improve—is compatible [19]. The added Bullet rule is that transport success, task closure, session state, and model judgment cannot confer mutation, verification, or integration authority.

DeepSeek’s session-log invariant—model-visible means logged—is the correct epistemology for a *cognitive* plane [13]. Bullet Farm needs the same discipline for prompts, tool schemas, and injected context. It must not use that log as repository truth. A reconstructable transcript proves what the model saw. It does not prove what landed on *main*.

Omnigent’s L7 credential proxy is the correct shape for forge and provider secrets: the sandbox never holds the real token; the proxy attaches it on approved egress [7], [16]. ADR 0007 and the provider-admission crate already strip host secrets and project minimum OAuth copies [51], [63]. The remaining work is audited provider-only egress, not a decision to put `GH_TOKEN` back in the child environment. Omnigent’s multi-device session sharing is a collaboration product. Bullet Farm’s Portal is the opposite product on purpose: a sequence-bound projection with no skip-checks affordance [55].

Typical single-agent harnesses remain the right *proposal producers*. V1 GA requires conformant Claude, Codex, Cursor, and Antigravity adapters under one proposal/evidence contract [18], [54]. Wrapping them (Omnigent) or replacing their loop (DeepSeek) is a different layer. Bullet Farm’s layer is the transaction around them.

B. Interoperability Without Authority Leakage

ACP and MCP are emerging interoperability layers [21], [27]. They should be consumed as capability observations and tool seams, never as mutation authority. A Cursor ACP session that streams a patch is a provider transcript. An MCP server that can run `git` is a hostile tool until it is reduced to a typed proposal or an admitted gate. The same rule applies to vendor “plan” modes: they are useful read-only observations. They do not admit a write.

X. IDENTITY, QUOTA, AND THE HUMAN SEAT

Gas Town’s scheduler is a rate governor (`scheduler.max_polecats`) aimed at 429s [5].

That is necessary and not sufficient. The historical Bullet Farm identity design—still designed, not a live quota plane—names profiles with owner, authorization class, custody mode, credential generation, and health [20], [48]. Personal CLI subscriptions stay in local custody; API and enterprise credentials use envelope encryption. Login success comes from an adapter `auth_status` probe, never from a UI button. Quota observations are typed and ranked: official API, structured event, headers, typed limit, estimator, then unknown. unknown is never scheduled as headroom. A bounded probe reservation may convert unknown to known without scheduling bulk work on a guess.

Named human subscriptions are rate-governed to one seat-equivalent per human regardless of how many consumer logins an operator can collect. Throughput must come from `api_credit`, `organization_seat`, or `service_identity`. That rule is a compliance-as-code stance, not legal advice, and it is not enforced in the current kernel. It is recorded here because it is the subsystem orchestrators most often omit and because it is part of why a fleet OS is not “just more tmux panes.”

XI. RESOLUTION MAP

Table V is how this paper “addresses” the failure classes without claiming they are all enforced in production. Status uses the closure vocabulary: component-complete, designed, or blocked [54].

Two adjudicated hardenings from the historical red-team remain designed, not shipped: wound-wait for multi-resource expansion, staged race budgets, and tree-disjoint evidence preservation [20]. Identifier renaming before review and blanket transitive locking were rejected as regressions. The paper adopts that discipline: a fix that destroys reviewability or serializes the repository is not a fix.

XII. EVALUATION AND FALSIFICATION

A fair comparison uses one repository/task corpus, one tool budget, one wall-clock limit, the same provider profiles, the same retry cap, and the same integration policy [19]. Report distributions, not a single success rate. Minimum metrics: in-window Candidate survival and protected-integration rate;

TABLE V
FAILURE CLASS TO BULLET FARM RULE. STATUS IS HONEST AGAINST
HEAD, NOT AGAINST THE SPECIFICATION.

Failure class	Rule	Status
Shared worktree / shared cwd	Ban worktrees; private clone per Attempt; <code>.git</code> file $\Rightarrow$ <code>WORKTREE_FORBIDDEN</code>	Component; production clone refused
-reference GC race	-dissociate (or CoW reflink) while mirror lock held	Dissociate implemented; reflink fast path designed
Hostile Git config / hooks	Clear <code>GIT_*</code> ; parse local config; refuse filters/aliases/includes	Component tests exist
Model <code>git</code> / inherited tokens	Provider read-only; empty credential helper; no remote; strip <code>GIT_*/GH_TOKEN</code>	Admission component; live spawn blocked
Control plane in <code>Git/Dolt</code>	SQL ledger for leases; BulletGit for Candidates; forge for refs	Ledger and journal components; not one transaction
<code>tmux</code> / silence as liveness	Lease + monotonic fence; four-valued observation; STONITH self-kill	Lease/fence component; full observation designed
Timeout-after-commit duplicate mint	Idempotent graph materialize; UNKNOWN never auto-retried	Command idempotency component; graph mint not a live path
Stale-green after rebase	Evidence names exact OIDs; move invalidates; tree-disjoint reuse is a whitelist	Fixture E2 only
Compile = code execution	Gates run in the writer/verifier sandbox class; network-denied [63]	Designed; one fixture gate
Cleanup by name/path	Preserve sealed generation; cleanup only by receipt	Local preservation component
Fleet as merge authority	Forge required checks; attestor $\neq$ broker	Designed; live forge blocked
Inferred empty/healthy UI	Watermarked projections; UNKNOWN $\neq$ empty success [55]	Component on projected surfaces

escaped-defect and revert rate; false-completion and zero-test rate; duplicate or ambiguous effect rate; crash-recovery time and lost-work bytes; spend, elapsed time, and human interventions; verifier backlog p50/p95.

No benchmark begins while synthetic proof substitutes for the five-plane transaction. No public comparison is published without raw task subjects, configuration, receipts, exclusions, and the exact upstream commits in Section IX. The first legal corpus is therefore: one Mission that materializes exactly once under kill-9 at every plane; one Variant that takes a never-reused fence; one private clone with no surviving `alternates`; one independent E2 reconstruction the author cannot weaken; one brokered effect whose timeout stays UNKNOWN until identity-exact read-back; and one portal snapshot that refuses to paint that UNKNOWN green [54], [61]. Until that saga has a signed receipt, Tables I and IV are architectural comparisons, not performance results.

The design is rejected—not “improved in a later version”—if any of the following is observed on an admitted build:

- a stale fence is accepted for a mutation, effect, or settlement;
- one command materializes two live graphs or two writers in one checkout;
- writer-produced proof satisfies an independent Evidence tier;
- UNKNOWN, timeout, zero tests, or a skipped gate is stored or rendered as VERIFIED;
- a worktree, shared cwd, or surviving `alternates` file is admitted as a writer workspace;
- the portal paints an unwatermarked or contradictory snapshot as a healthy empty list.

Those conditions are executable. They are the opposite of a scoreboard.

XIII. WHAT THIS PAPER DOES NOT CLAIM

A red team that needs a list of forbidden promotions can stop here.

- We do not claim measured superiority on latency, cost, autonomy, or defect escape against Gas Town, Gas City, DeepSeek Harness, Omnigent, or any single-agent harness.
- We do not claim a connected five-plane TRANSACTION_PROOF at HEAD, a hub-only installer, or a signed five-target release [61].
- We do not claim live Claude, Codex, Cursor, or Antigravity conformance. Offline parsers and refused spawn are not live receipts.
- We do not claim that the archived 2026-08-24 live demo certifies the current tree.
- We do not claim exactly-once physical side effects. Ambiguous remote writes stay UNKNOWN until identity-exact adoption.
- We do not claim that evolutionary search, PostgreSQL, or remote runners exist beyond design documents and refused scaffolds.
- We do not claim Jankurai, SBOM, or supply-chain release quality.
- We do not claim that a green just fast or a green crate test is product readiness.
- We do not claim that finishing this preprint, or writing the product-gap register, closed G1–G15.

If a sentence in marketing copy contradicts this list, the list wins.

Artifact and reproducibility statement

The source of this preprint is the hub path `docs/paper/` in the public Bullet Farm family. It is compiled with `IEEEtran` and the companion `refs.bib`. It is not a family release subject. Competing systems are pinned in Appendix A and in the dated competitor snapshot [19]. A reader who wants to reproduce the *inventory* must replay `bullet-family check release -json` and the mapped component tests on the named commits, not trust this PDF. A reader who wants to reproduce a *benchmark* cannot: none is claimed.

XIV. RELATED WORK

Transaction processing is the ancestor [46], [64]: identity, isolation, and recovery under crash. Sagas name why a long-running remote effect cannot be a local commit [65]. Confused-deputy and capability discipline are why a portal, MCP server, or vendor sandbox must not inherit forge credentials [66]. This paper applies those ancestors to model-authored software changes.

On the coding-agent side, SWE-bench is the dominant issue-resolution benchmark [28]; SWE-agent, Agentless, OpenHands, and AutoCodeRover optimize an agent–computer interface or a no-agent scaffold for that benchmark [30]–[32],

[67]. Multi-agent role graphs (ChatDev, MetaGPT, AutoGen) assign personas, not incarnation-fenced write authority [68]–[70]. Prompt injection through retrieved or repository-resident instructions is a documented confused-deputy class [58]. None of those systems bind Evidence to an exact Candidate OID.

Jujutsu shows why a stable Change identity matters and why a Git trailer is not enough [47]. Bors-style merge queues show why landing must be serialized and bisected [38]. in-toto and SLSA are the supply-chain vocabulary for the release artifacts Gas City already ships and Bullet Farm does not [71], [72]. Linux isolation neighbors are bubblewrap, Landlock, and credential-proxying meta-harnesses [16]. ACP and MCP are transport and tool seams [21], [27]. Gas Town, DeepSeek Harness, and Omnigent are the nearest *product* neighbors and are treated as such in Sections II and IX, not as straw men.

XV. LIMITATIONS

Bullet Farm is pre-release [50], [61]. There is no connected TRANSACTION_PROOF at HEAD. Live provider, Jeryu, and GitHub gates are operator-blocked. Hub-only installation is refused because the checked-in lock is schema 2. Security quality is below the V1 bar (Jankurai 58, 26 hard findings). Eleven of fifteen Portal surfaces stay explicit UNKNOWN. Evolutionary multi-agent search is a normative document, not a running optimizer [48]. C2, C3, C4, C5, C6, C6b, C9–C12 are designed or out of V1; only C7 is a complete formal-methods receipt. PostgreSQL, remote runners, and cross-repository sagas are scaffolds or absences. This preprint’s authors are placeholders. Comparison cells for DeepSeek and Omnigent Git internals that are not stated in their public architecture are marked unknown rather than inferred from plugin names. The inventory HEADs in Section VIII are a dated snapshot; the working tree that compiled this PDF may be newer.

XVI. CONCLUSION

The coding-agent industry is composing loops. Bullet Farm is a transaction processor for the thing those loops actually mutate: repositories and forges, under crash, timeout, and adversarial concurrency. Gas Town showed that roles, convoys, and a merge queue are valuable and that tmux, worktrees, and inferred completion are a ceiling. DeepSeek Harness showed that a session log and a plugin kernel make a single agent inspectable and recomposable. Omnigent showed that OS isolation, credential proxying, and cross-harness policy can be layered on top of the agents people already run. Bullet Farm keeps those product lessons and changes the substrate: private clones, incarnation fences, exact Candidates, independent Evidence, brokered effects, and a portal that would rather show UNKNOWN than a false empty success.

That is why the architecture is different. The components that exist today already enforce pieces of that contract and refuse the rest. The connected product is not done until the receipt table says so.

APPENDIX A OBSERVATION PINS

All moving-repository observations in this paper are dated 2026-08-25 UTC.

- Gas Town: release v1.2.1, peeled commit 319d33a91b2deca59bba6dd26be6b9daf8eaacf6, observed 2026-08-25.
- Gas City: release v1.4.1, peeled commit 58ef17e3bd685fd5cf7f21286277b208d3324590; development main 2cd07e018bf3680d24b037b509e6a4bad5
- DeepSeek Harness: tag dsh-v0.1.1-rc.2, commit b150a551b8d465e31e418e1b2eaf5e79bbb7d28e.
- Omnigent: release v0.10.0, commit 40755dd8dddb07e1eb6e4055d1d9936e184ceb9b; development main 00bbbed5d92d628f975053e82dbfd1ecd54
- Bullet Farm family HEADs used for the inventory: hub bf5c64245d218050eb8f4c3228ea15654540a94e, kernel 0346fd531e231a88e5b1eb3a8257be04f235068b, BulletGit 236f4efe2b8243b3c4dd5da13269e45119b061df, portal 8272844e44d89ea3b26aa5b3c4f25909a069e101 [54]

A later comparison must record a new tag or commit and date. Silently tracking main is forbidden [19].

APPENDIX B RED-TEAM SELF-REVIEW

A hostile reviewer will run the following attacks. We accept each one and state the residual.

- “*You compared yourselves to a cartoon of Gas Town.*” Residual accepted if we had omitted Refinery, Seance, Wasteland, or Gas City 1.4.1 artifacts. Those are credited in Section II. The remaining claim is about bindings, not about missing roles.
- “*DeepSeek’s session log is evidence.*” Residual accepted as a cognitive-plane truth. Rejected as repository truth. A reconstructable transcript is E0/E1 input, not E2.
- “*Omnigent’s sandbox plus credential proxy is stricter than your V1.*” Residual accepted for OS isolation maturity: Omnigent ships wrappers today. Rejected as write-authority: the wrapped harness still runs Git. Our Linux egress component is live-tested and policy-gated [56].
- “*Your implemented row is empty, so the architecture is vapor.*” Residual accepted for the connected product. Rejected for named component receipts. Table III is the contract with the reader.
- “*#5511 is fixed upstream, so the autopsy is stale.*” Residual accepted as a moving-issue risk. The class—timeout after server-side commit plus auto-retry—remains a missing binding wherever effect identity is not exact. We pin the issue URL and date rather than claiming it is still reproducible on main.
- “*Two TLA+ models prove the system.*” Residual accepted: they prove two protocols, and only if the model includes the conflict path [20]. They do not prove the product.
- “*An internal score is a result.*” Residual rejected. This revision deletes the unpublished 26/100 figure. The citable objects are receipt rows and blocked gates.
- “*You will enable live admission in a doc.*” Residual rejected. ADR 0012 is Proposed; committed policy stays generation 1 and live-false [57], [61].
- “*A worktree with git worktree add and a unique branch is isolation.*” Residual rejected. Shared .git, shared hooks, and a .git file remain [43], [45].

- “*C1–C12 were silently dropped.*” Residual rejected. Section VI and Table II are the disposition. Designed is not enforced.
- “*Closing the white paper closed the product.*” Residual rejected. Appendix C is a visibility close. G1–G15 remain implementation, ratification, and receipt work [61].
- “*The eleven UNKNOWN portal surfaces are unfinished UI.*” Residual rejected as a honesty bug. They are the correct projection of a missing farmd route. Painting them empty-success would be the defect.
- “*v1alpha2 on HEAD means live is on.*” Residual rejected. The committed policy stays v1alpha1, generation 1, live_admission_enabled=false. The v1alpha2 artifact is a fixture and a validator. The Kernel snapshot loader still accepts v1alpha1 only [57].
- “*Your inventory HEADs are yesterday’s tips.*” Residual accepted for any later commit. This revision pins the later 2026-08-25 closure subjects named in Appendix A. Earlier preprint drafts named hub 74507b3 / kernel ba70d8b / BulletGit 5dac98e / portal d607068; those are superseded as inventory subjects.
- “*Same-UID swap after verify is a hidden install hole.*” Residual accepted as G1. Schema-2 source setup cannot close it. A signed prebuilt installer is the closer [61].

APPENDIX C REMAINING PRODUCT GAPS

Table VI is the operator-facing remainder. The executable form is bullet-family check release -json (26 static negative gates). The prose form is the hub product-gap register, which maps every release.*id onto G1–G15 [61]. This appendix exists so a red team cannot say the paper hid an open gate behind architecture.

Operator-only closers, which no agent may flip: ratify ADR 0012 with a real runner key; restore scoped Jeryu authentication on an unmodified forge; publish schema-3 lock inputs; configure a GitHub App test repository. Documentation that waits on typed commands—upgrade/rollback, signer rotation, freeze-enforced cards—is deferred on purpose. Inventing those runbooks now would invent a CLI.

V1 is done when every closure-plan slice V1–S0..V1–S8 has a current independently verifiable receipt from the same signed subjects [54]. “Agents ran” is not that receipt.

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TABLE VI
REMAINING V1 PRODUCT GAPS (2026-08-25). NONE OF THESE IS CLOSED BY THIS COMPILE.

ID	Gap	Why it is still open	What already exists
G1	Hub-only signed install	Schema-2 lock refused on purpose	Descriptor-relative setup fixture
G2	Five-plane transaction	No signed TRANSACTION_PROOF	Atomic lease/command; fixture E2
G3	Production Kernel writes	No signed lease transport / JSON-RPC dispatch	DB-clock leases; launch grant
G4	Production BulletGit writes	Public clone fail-closed	Dissociate, hostile-git, preservation
G5	Live provider conformance	Committed policy live-false	Four offline parsers; egress probes
G6	Jeryu live effect	No authenticated read-back	Local bare-forge component
G7	GitHub live effect	No App-test-repo receipt	Specified adapter
G8	Security release floor	Jankurai 58; 26 hard findings	Fail-closed local scan
G9	Signed five-target release	No builder / SBOM / protected signer	Verify+extract of a signed archive
G10	Non-Linux containment	Mutation refused off Linux	Linux isolation reference
G11	Evolutionary runtime	Recipes are design-only	Normative evolutionary document
G12	Family check release	26/26 gates BLOCKED	Negative inventory
G13	Portal product surfaces	11 of 15 surfaces explicit UNKNOWN	Four projected surfaces; CSRF/202
G14	farmd production API	Snapshot/SSE subset, not ~80 routes	Authenticated command ingress
G15	Cognitive persistence	Tasks/roles are IDs or design	Wire shapes; offline parsers

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